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Case 29-2025: A 43-Year-Old Woman with Depression, Suicidal Ideation, and Fever

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PRESENTATION OF CASE

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CME



Dr. Heidi S. Park (Medicine): A 43-year-old woman was admitted to the psychiatry service of this hospital because of depressed mood and suicidal ideation.

The patient had been in her usual state of health until 2 months before the current admission, when she began to have worsening anhedonia, anxiety, and suicidal ideation, including plans to intentionally ingest an overdose of medication. On the day of the current admission, the patient presented to the emergency department of this hospital for evaluation. She reported 2 months of nonproductive cough without night sweats or weight loss. Multiple psychosocial stressors were present, including intimate partner violence, limited financial resources, and unstable housing. The patient was admitted to the psychiatry service of this hospital for management of anhedonia and suicidality.

On evaluation, the temporal temperature was 37.3°C, the blood pressure 90/56 mm Hg, the pulse 70 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 99% while the patient was breathing ambient air. She was mildly somnolent and diaphoretic. No conjunctival hemorrhages, oropharyngeal plaques, or lesions on the mucous membranes were noted. The cardiopulmonary examination was normal, with clear lung fields and no murmur. No palpable cervical, supraclavicular, axillary, or inguinal lymphadenopathy was present. No hepatosplenomegaly was detected on palpation of the abdomen, and the arms and legs had no rash, edema, or signs of embolic disease.

The patient's medical history was notable for human immunodeficiency virus (HIV) infection with associated neuropathy, which she had acquired 17 years earlier; adherence to antiretroviral medications had been limited in the 2 years preceding this evaluation. The patient had a history of bipolar disorder, as well as seizure disorder that was secondary to traumatic brain injury. She did not take any medications but had previously received a prescription for combination antiretroviral therapy (ART) with bictegravir, tenofovir alafenamide, and emtricitabine. Her surgical history was notable for Roux-en-Y gastric bypass that had been performed 10 years earlier. The patient had unstable housing; she had stayed with friends in Boston but had not sought temporary housing in shelters. She smoked cocaine as well as one pack of cigarettes per day,

and she drank alcohol recreationally. She had not traveled outside New England for several years.

The CD4+ T-cell count was 25 per microliter (reference range, 295 to 1471), and the HIV RNA viral load was 144,000 copies per milliliter (assay range, 20 to 10,000,000). The lactate dehydrogenase level was 227 U per liter (reference range, 110 to 210). Urinalysis showed no evidence of pyuria, hematuria, or proteinuria. Other laboratory test results are shown in Table 1.

On the evening of the first hospital day, the oral temperature was 39.5°C, the blood pressure

83/54 mm Hg, the pulse 101 beats per minute, the respiratory rate 40 breaths per minute, and the oxygen saturation 94% while the patient was breathing ambient air. She had an ashen appearance, and increased work of breathing was noted.

Dr. Robert J. Lawson: An anteroposterior chest radiograph (Fig. 1) revealed innumerable micronodules that were evenly distributed throughout both lungs.

Dr. Park: Lactated Ringer's solution was administered intravenously, along with ceftriaxone, doxycycline, and trimethoprim-sulfamethoxazole.

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Initial Presentation
Hemoglobin (g/dl)	13.5–17.5	10.7
Hematocrit (%)	41.0–53.0	34.0
White-cell count (per μ l)	4500–11,000	8630
Differential count (per μ l)		
Neutrophils	1800–7700	7580
Lymphocytes	1000–4800	350
Monocytes	200–1200	560
Eosinophils	0–900	20
Basophils	0–300	30
Immature granulocytes	0–100	90
Platelet count (per μ l)	150,000–400,000	426,000
Sodium (mmol/liter)	135–145	136
Potassium (mmol/liter)	3.4–5.0	2.9
Chloride (mmol/liter)	98–108	97
Carbon dioxide (mmol/liter)	23–32	25
Urea nitrogen (mg/dl)	8–25	14
Creatinine (mg/dl)	0.60–1.50	0.6
Calcium (mg/dl)	8.5–10.5	8.6
Aspartate aminotransferase (U/liter)	10–40	22
Alanine aminotransferase (U/liter)	10–55	11
Alkaline phosphatase (U/liter)	15–115	221
Total bilirubin (mg/dl)	<1.2	0.2
Albumin (g/dl)	3.3–5.0	2.4
Globulin (g/dl)	1.9–4.1	3.9
Lactate dehydrogenase (U/liter)	110–210	227
Creatine kinase (U/liter)	40–150	56

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

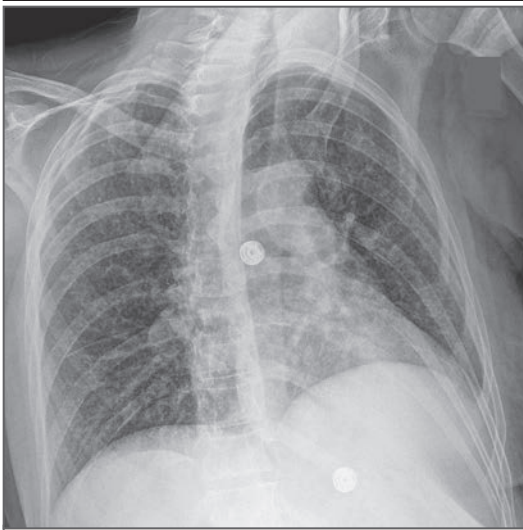


Figure 1. Chest Radiograph on Presentation.

An anteroposterior image obtained on portable chest radiography shows pulmonary nodules in a miliary distribution.

The patient was transferred to the medical service for further treatment.

Diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Michael S. Abers: This 43-year-old woman with advanced HIV infection and bipolar disorder presented with 2 months of anhedonia, suicidal ideation, and cough. Immunocompromised patients are susceptible to a variety of infections, cancer, and disorders of immune dysregulation. The risk of a particular disease depends on the nature and severity of the underlying immunologic impairment, as well as epidemiologic factors. When formulating a differential diagnosis for this patient, I will address the following three questions: What is the nature and severity of this patient's underlying immunologic defect? Does she have epidemiologic factors that warrant special consideration? Are there any salient clinical features that would help narrow the differential diagnosis?

UNDERLYING IMMUNOLOGIC DEFECT

What is the nature and severity of this patient's underlying immunologic defect? HIV infection causes various host defense impairments.¹ In ad-

dition to the defects in cellular immunity caused by CD4⁺ T-cell lymphopenia, HIV infection leads to dysregulation of humoral immunity and innate immunity. This patient presented with advanced immunodeficiency that was associated with vulnerability to the entire spectrum of illnesses related to acquired immunodeficiency syndrome (AIDS).

EPIDEMIOLOGIC FACTORS

Does the patient have epidemiologic factors that warrant special consideration? Susceptibility to most AIDS-related illnesses depends not only on the degree of immunocompromise but also on epidemiologic factors. Many opportunistic infections are endemic to specific geographic regions. The patient was asked about travel history, as well as current and previous locations of residence, and she reported no history of travel to or previous residence in geographic regions outside New England. Her history of unstable housing increases her risk of certain infections, including *Mycobacterium tuberculosis* infection and *Bartonella quintana* infection. Cigarette smoking confers a predisposition to respiratory infections, primary lung cancer, and other cancers. Smoking or insufflating cocaine can result in inhalational lung injuries and pulmonary hemorrhage.

SALIENT CLINICAL FEATURES

Are there any salient clinical features that would help narrow the differential diagnosis? Although patients with advanced HIV infection are at risk for an extensive list of pulmonary conditions, five aspects of this patient's clinical presentation help narrow the differential diagnosis.

First, the patient reported a 2-month history of cough, a feature consistent with a subacute or chronic tempo of illness. This duration of symptoms effectively rules out viral lung infections as well as infections with most community-acquired bacterial pathogens (*Streptococcus pneumoniae*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, and legionella), which have a more acute presentation. Second, chest imaging showed diffuse micronodular opacities that were evenly distributed throughout both lungs, which is consistent with a miliary pattern. Miliary nodules, named for their resemblance to millet seeds,² arise from the hematogenous spread of underlying disease, which is typically granulomatous or malignant in nature.

Third, routine blood testing revealed an elevated alkaline phosphatase level in the presence of normal levels of aspartate aminotransferase, alanine aminotransferase, and bilirubin, findings that suggest the possibility of an infiltrative liver disorder and thereby invoke a multisystem process. Fourth, the lactate dehydrogenase level was only mildly elevated, below the level typically seen in patients with pneumocystis pneumonia, disseminated histoplasmosis, or non-Hodgkin's lymphoma. Fifth, hematologic studies revealed thrombocytosis. It is notable that most opportunistic infections either have no effect on the platelet count or result in thrombocytopenia. In contrast, tuberculosis and certain cancers can be associated with thrombocytosis.

When constructing the differential diagnosis for this patient, it is important to consider the fact that profoundly immunocompromised patients frequently present with multiple concurrent opportunistic conditions. Thus, the principle of Occam's razor need not apply. Nonetheless, I will attempt to identify a single diagnosis that accounts for all the salient features in this patient's case.

NONINFECTIOUS IMMUNE-MEDIATED CONDITIONS

The differential diagnosis of miliary lung nodules includes several noninfectious immune-mediated conditions, such as occupational pneumoconiosis, foreign-body granulomatosis, hypersensitivity pneumonitis, and sarcoidosis. The absence of exposures in this case rules out diagnoses associated with inhalation or insufflation. Sarcoidosis is a rare cause of miliary lung nodules. Although the pathogenesis of sarcoidosis is poorly understood, CD4+ T cells appear to play a critical role. It is interesting to note that CD4+ T-cell lymphopenia in patients with HIV infection confers a reduced risk of incident sarcoidosis.³ Overall, noninfectious immune-mediated disease is a highly unlikely diagnosis in this patient.

CANCER

Patients with advanced HIV infection are at risk for several cancers that may manifest with lung involvement, including Kaposi's sarcoma, non-Hodgkin's lymphoma, and primary lung cancer. Kaposi's sarcoma is associated with mucocutaneous lesions, male sex, and a characteristic radiographic appearance, features that were absent in this patient. The absence of these findings

rules out the diagnosis. In patients with advanced HIV infection, non-Hodgkin's lymphoma often manifests with cytopenias affecting multiple cell lines, a markedly elevated lactate dehydrogenase level, pleural effusions, and hilar lymphadenopathy, all of which were absent in this patient; lung involvement is rare with this condition.^{4,5} Among patients with HIV infection, the risk of primary lung cancer is 2 to 3 times that in the general population. The disease can occur at any CD4+ T-cell count and is especially likely to develop in persons who smoke. Miliary involvement is a rare but well-described manifestation of lung adenocarcinoma.⁶ Thrombocytosis is relatively common in patients with lung cancer, occurring in one fourth of cases at the time of the initial diagnosis, whereas it is uncommon with most opportunistic infections.⁷ In most patients with lung cancer, a dominant nodule can be identified on imaging. Although cancer cannot be ruled out in this patient, an infectious cause is more likely to account for her presentation.

ENDEMIC MYCOSES

Endemic dimorphic fungi — including histoplasma, blastomyces, coccidioides, and paracoccidioides — are well-established causes of miliary lung nodules. Histoplasma is endemic in Central America and the Ohio and Mississippi River valleys in the United States. In these regions, disseminated histoplasmosis is a common opportunistic infection among persons with AIDS and is a well-described cause of miliary lung nodules.^{8,9} Disseminated infection with multi-system involvement is common. Characteristic laboratory findings include pancytopenia and an extremely elevated lactate dehydrogenase level (often >1000 U per liter).¹⁰ Although histoplasma is not endemic in New England, environmental modeling studies suggest an expanding geographic region of endemicity, and cases are increasingly being reported in regions that are not typical for histoplasmosis.¹¹ Ultimately, the clinical, epidemiologic, and laboratory manifestations in this case do not support the diagnosis of histoplasmosis. Similarly, coccidioidomycosis, blastomycosis, and paracoccidioidomycosis can be ruled out in the absence of a relevant exposure history, given that these infections are more geographically restricted.

UBIQUITOUS MYCOSES

Pneumocystis jirovecii infection is the most common fungal respiratory infection in HIV-positive patients with a CD4+ T-cell count of less than 200 per microliter.¹² Patients with this infection typically present with the subacute onset of fever, nonproductive cough, dyspnea, and hypoxemia. On chest radiography, the classic finding is diffuse, bilateral interstitial opacities. Miliary nodules can be seen but are less common.^{8,13} The lactate dehydrogenase level is elevated in nearly all cases, and levels greater than 300 U per liter are present in 75% of cases.¹⁴ Anemia and hypoalbuminemia are common, but thrombocytosis and elevated alkaline phosphatase levels are not associated with this infection. The clinical, laboratory, and imaging manifestations in this case do not support the diagnosis of pneumocystis pneumonia, but given the prevalence of the infection in patients with advanced HIV infection, it is important to consider the possibility of an uncommon manifestation of this common illness.

Cryptococcus neoformans infection should also be considered in the differential diagnosis. *C. neoformans* infection is common in patients with advanced HIV infection and primarily occurs when the CD4+ T-cell count falls below 100 per microliter. Although isolated pulmonary syndromes have been reported, meningoencephalitis is present in more than 90% of cases. Furthermore, miliary lung nodules rarely occur in patients with pulmonary cryptococcosis. Thus, this infection is an unlikely explanation for the patient's clinical presentation.

NONTUBERCULOUS MYCOBACTERIAL INFECTIONS

Nontuberculous mycobacterial infections should be considered in the differential diagnosis. The most common nontuberculous mycobacterial infection in HIV-positive patients is disseminated *M. avium* complex infection. *M. avium* complex is weakly pathogenic. The infection occurs when the CD4+ T-cell count falls below 50 per microliter and is usually disseminated, as opposed to the localized syndromes seen in immunocompetent persons. Disseminated *M. avium* complex infection manifests with constitutional symptoms, which are often accompanied by abdominal pain, diarrhea, and hepatosplenomegaly. Patients often have anemia, hypoalbuminemia, and elevated alkaline phosphatase levels; pancytopenia is usu-

ally present. Moreover, although the organisms may be detected in respiratory secretions, clinically significant lung disease is distinctly uncommon. For these reasons, *M. avium* complex infection can be ruled out as the cause of this patient's respiratory illness.

Patients with HIV infection are also at risk for *M. kansasii* infection. Unlike disseminated *M. avium* complex infection, *M. kansasii* infection typically involves the lung in HIV-positive patients, and clinical disease may be indistinguishable from tuberculosis. Extrapulmonary dissemination is seen in approximately one fourth of patients, and miliary disease has been reported.¹⁵ Although *M. kansasii* infection cannot be ruled out in this case, the incidence of *M. kansasii* infection is substantially lower than that of tuberculosis among patients with HIV infection, and the clinical manifestations are more consistent with a diagnosis of miliary tuberculosis caused by disseminated *M. tuberculosis* infection.

MYCOBACTERIUM TUBERCULOSIS INFECTION

Given this patient's clinical presentation, I strongly suspect that the underlying diagnosis is miliary tuberculosis. Patients with impairments in cellular immunity and other host defense defects that impair granuloma integrity are at an especially high risk for reactivation of *M. tuberculosis* infection and for a disseminated or progressive primary illness at the time of initial infection with the pathogen.

The risk of reactivated tuberculosis begins to increase very early during HIV infection, before the decline in the CD4+ T-cell count in peripheral blood. Persons with a CD4+ T-cell count below 200 per microliter present with atypical manifestations of pulmonary tuberculosis, such as involvement of the middle and lower lung fields, intrathoracic lymphadenopathy, and pleural effusions. Hemoptysis and cavitary lesions are uncommon.^{16,17} In addition, advanced HIV infection confers a strong predisposition to extrapulmonary dissemination, which may be present in two thirds of cases.¹⁸ Miliary tuberculosis is particularly common among patients with HIV infection and accounts for half of cases of extrapulmonary tuberculosis in this patient population.

Miliary tuberculosis typically manifests with a subacute or chronic febrile respiratory illness.¹⁹ Despite widespread involvement of infection out-

side the lung, localizing signs and symptoms are typically absent. A wide range of laboratory abnormalities can be seen in patients with miliary tuberculosis. Anemia, hypoalbuminemia, and hyponatremia, as well as an elevated alkaline phosphatase level with normal levels of bilirubin, aspartate aminotransferase, and alanine aminotransferase, are particularly common. The lactate dehydrogenase level may be mildly elevated in patients with miliary tuberculosis, but extremely high levels (such as those seen with histoplasmosis and pneumocystis pneumonia) are uncommon.

In most cases, miliary nodules can be detected on chest radiography on the initial presentation. Computed tomography (CT) of the chest allows for further characterization while providing enhanced sensitivity for the detection of intralobular septal thickening, lymphadenopathy, and pleural effusions. Although any organ can be involved, the most common sites of extrapulmonary dissemination include the lymph nodes, liver, bone marrow, and central nervous system (CNS). In addition to the lung and probable liver involvement in this patient, she had mild lethargy on presentation that could conceivably reflect early evidence of CNS involvement. Both tuberculosis and HIV infection have been associated with a wide range of neurologic syndromes. Tuberculomas and basilar meningitis are the most common with tuberculosis; HIV-associated neurocognitive disorder (HAND) is the most common with HIV infection.

Could CNS tuberculosis explain the depression and suicidality that prompted this patient's initial presentation? Although brain tuberculomas can cause psychiatric symptoms, alternative explanations are more likely. First, we must consider the contribution of social determinants of health. This patient had multiple severe psychosocial challenges, including intimate partner violence, financial insecurity, unstable housing, and substance use. Collectively, these issues could certainly trigger a depressive episode, especially in a patient with underlying bipolar disorder. Tuberculosis could be a contributing or exacerbating factor in the patient's psychiatric illness. Critical to the long-term care of this patient is a multidisciplinary approach that addresses both psychiatric and medical diagnoses. Further exploration of the patient's nonadherence to ART is critical to preventing future infections.

In patients with HIV infection, establishing the diagnosis of tuberculosis, particularly miliary tuberculosis, can be challenging. HIV-positive patients with miliary tuberculosis have fewer cavitory lesions and a lower organism burden in the lung than HIV-negative patients with pulmonary tuberculosis, which diminishes the sensitivity of acid-fast staining and nucleic acid amplification testing of sputum samples. Given the extent of this patient's lung disease, the first step in establishing the diagnosis is to obtain sputum specimens for a mycobacterial smear, culture, and nucleic acid amplification testing.

DR. MICHAEL S. ABERS' DIAGNOSIS

Miliary tuberculosis.

ADDITIONAL IMAGING STUDIES

Dr. Lawson: On the second hospital day, CT of the chest was performed after the administration of intravenous iodinated contrast material (Fig. 2A). The images confirmed the previous radiographic findings of innumerable nodules (measuring 1 to 3 mm in diameter) in an even distribution throughout both lungs. An additional enlarged upper paratracheal lymph node with central low attenuation and a thickened rim was present. No mass, consolidation, cavity, or effusion was noted.

One day later, CT of the abdomen and pelvis was performed with the administration of oral and intravenous contrast material in the portal venous phase (Fig. 2B and 2C). Multiple retroperitoneal and upper abdominal lymph nodes had an appearance that was similar in character to the upper paratracheal lymph node. There were two low-attenuation lesions (measuring up to 4.1 cm in greatest dimension) in the liver that had thickened rims and some internal septation. There was an additional lesion (measuring 2.0 cm in greatest dimension) in the pancreatic head that had no features of pancreatic or biliary duct obstruction.

Magnetic resonance imaging (MRI) of the head was performed after the administration of gadolinium-based contrast material (Fig. 2D). A single subtle 3-mm focus of enhancement in the right inferior temporal subcortical white matter with minimal surrounding hyperintensity was present on T2-weighted imaging and on fluid-attenuated inversion recovery imaging. No restricted diffusion or susceptibility was present.

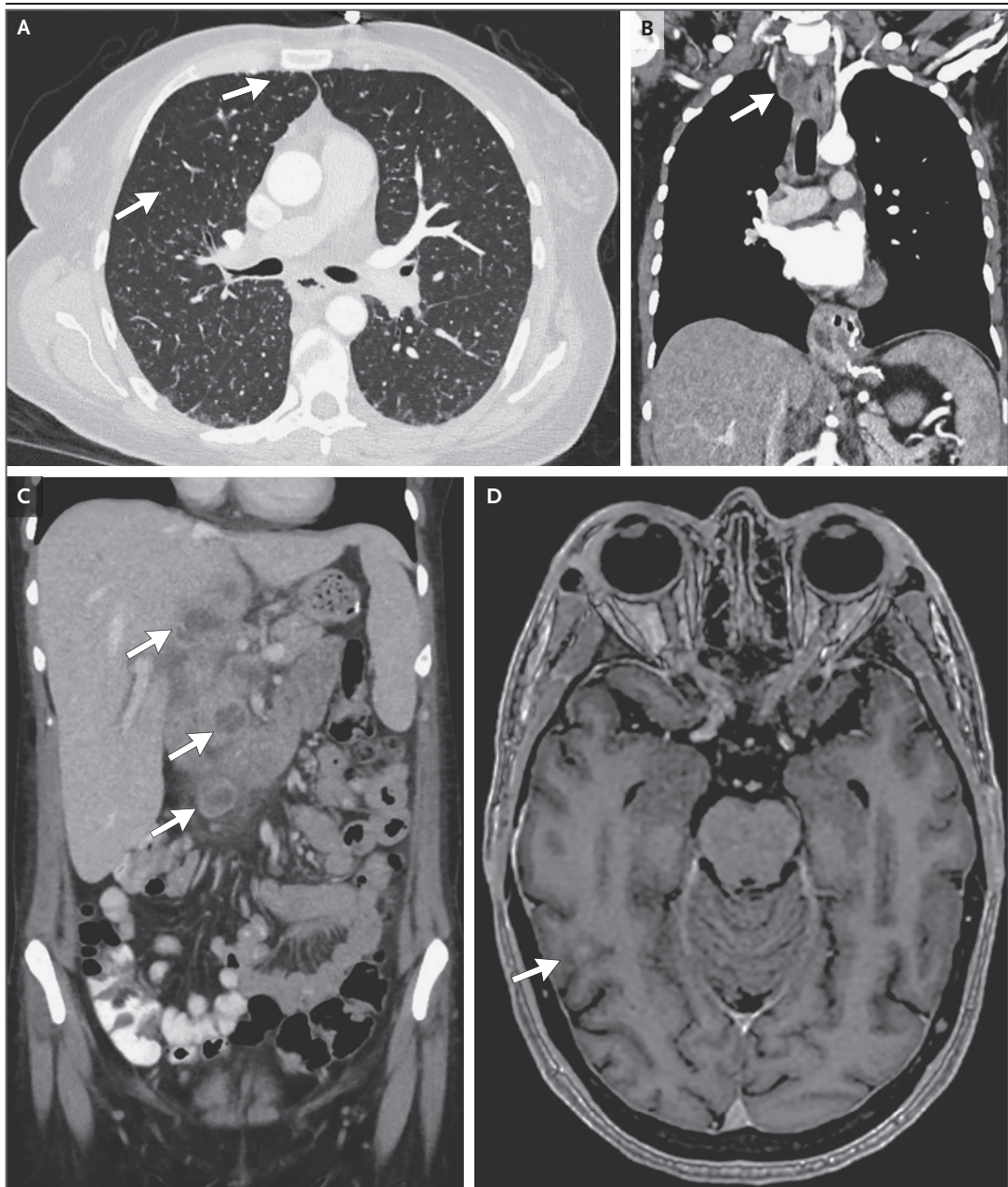


Figure 2. Additional Imaging Studies of the Chest, Abdomen, and Head.

CT of the chest was performed after the administration of intravenous contrast material, and the lung kernel was used as the image-reconstruction algorithm. An axial image (Panel A) shows nodules measuring 1 to 3 mm in diameter that are evenly distributed throughout both lungs (arrows), which is consistent with a miliary distribution. A coronal reformatted image in a soft-tissue window (Panel B) shows an enlarged right upper paratracheal lymph node with central low attenuation and a thickened rim (arrow). CT of the abdomen and pelvis was performed with the administration of oral and intravenous contrast material in the portal venous phase. A coronal reformatted image (Panel C) shows a lesion in the liver with central low attenuation and a thickened rim (top arrow), a similar lesion in the pancreatic head (middle arrow), and a necrotic upper mesenteric lymph node (bottom arrow). Multiple enlarged retroperitoneal lymph nodes with central low attenuation are also present. MRI of the head was performed after the administration of gadolinium-based contrast material. A T1-weighted image (Panel D) shows a single subtle 3-mm focus of enhancement (arrow) in the right temporo-occipital subcortical white matter.

PATHOLOGICAL DISCUSSION

Dr. Melanie C. Kwan: To expedite the diagnosis, induced sputum specimens were obtained, and bronchoscopy with transbronchial biopsy was performed. Staining of the induced sputum specimen was positive for acid-fast bacilli (Fig. 3A). The detection of acid-fast bacilli on a smear alone is insufficient to establish a definitive diagnosis of *M. tuberculosis* infection, and subsequent confirmatory nucleic acid testing was performed, in addition to culture.

A transbronchial-biopsy specimen from a paratracheal lymph node that was obtained during bronchoscopy showed fragments of histiocytic aggregates with palisading and associated acute inflammation, findings suggestive of necrotizing granulomas (Fig. 3B). Special staining for acid-fast bacteria showed abundant slender bacilli that had morphologic features consistent with *M. tuberculosis* complex (Fig. 3C). After 9 weeks, cultures were positive for *M. tuberculosis*. In addition, nucleic acid amplification testing performed by the Massachusetts State Public Health Laboratory confirmed the diagnosis of *M. tuberculosis*. Antimicrobial susceptibility testing did not detect any antimicrobial resistance.

PATHOLOGICAL DIAGNOSIS

Mycobacterium tuberculosis infection.

MANAGEMENT AND HOSPITAL COURSE

Dr. Sandra B. Nelson: This young woman has two life-threatening infections — advanced HIV infection and disseminated *M. tuberculosis* infection — both of which require urgent management. Furthermore, with advanced HIV infection, the known opportunistic infections are often just the tip of the iceberg; vigilance to the possible unmasking of other infections is necessary as treatment for the known diagnoses is initiated. The management of HIV and tuberculosis coinfection is also complicated by challenging drug interactions, polypharmacy, and the risk of immune reconstitution inflammatory syndrome (IRIS), a condition caused by an enhanced inflammatory reaction against tuberculous antigens in the context of immune recovery.

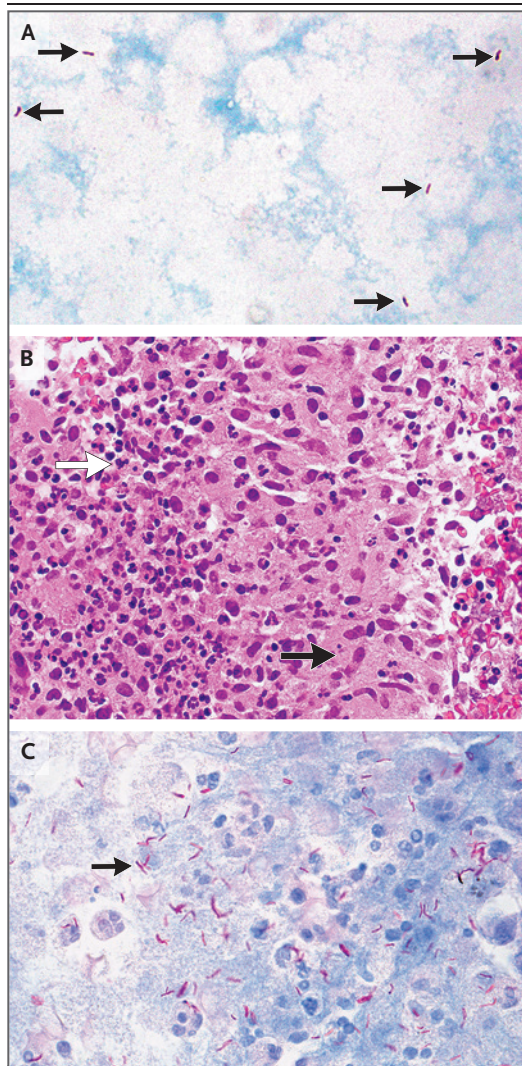


Figure 3. Sputum and Lymph Node–Biopsy Specimens.

Acid-fast staining of a sputum specimen (Panel A) highlights slender bacilli (arrows), a finding that is consistent with mycobacterium species but is nonspecific. Hematoxylin and eosin staining of a transbronchial-biopsy specimen from a paratracheal lymph node (Panel B) shows histiocytic aggregates with palisading (black arrow, indicating a representative histiocyte) admixed with acute inflammation and nuclear debris (white arrow), findings consistent with necrotizing granulomas. Acid-fast staining of the transbronchial-biopsy specimen (Panel C) shows numerous slender forms (arrow) located in the areas suggestive of necrotizing granulomas that had morphologic features consistent with *Mycobacterium tuberculosis* complex.

In a patient with HIV and tuberculosis coinfection, initial management priorities include ensuring that sufficient diagnostic material is sent for

culture, thereby optimizing the likelihood that results of antimicrobial susceptibility testing will ultimately be available; rapidly initiating treatment for tuberculosis; and determining the likelihood of tuberculous meningitis, which would inform decisions about antituberculous therapy and the timing of initiation of ART. Most patients with tuberculosis receive conventional empirical treatment with a four-drug regimen (rifampin, isoniazid, pyrazinamide, and ethambutol) while awaiting the results of antimicrobial susceptibility testing, barring any concern about resistance. The presence of tuberculous meningitis is an indication for glucocorticoid treatment and informs the dosing of antituberculous therapy for CNS penetration.

In this patient, there were no known risk factors for resistant tuberculosis, and therefore conventional four-drug therapy was initiated. Dexamethasone was added while the evaluation for tuberculous meningitis was ongoing. Lumbar puncture was performed. On cerebrospinal fluid (CSF) analysis, the nucleated-cell count was 1 per microliter, and the protein and glucose levels were normal. Ultimately, on the basis of the patient's clinical features, the absence of leptomeningeal findings on MRI, and the normal CSF analysis, tuberculous meningitis was thought to be an unlikely diagnosis. The presence of a small intracranial tuberculoma could not be ruled out, and moxifloxacin was added to her antituberculous regimen to optimize CNS penetration.

The next step was to select an ART regimen and determine the timing of treatment initiation. In any patient with HIV infection, the selection of ART is based on potency in restoring immune function, ease of administration, treatment history, antiretroviral resistance, and drug interactions. The rapid initiation of ART is a priority in patients with advanced HIV infection and has been shown to be lifesaving.^{20,21} However, patients with HIV and tuberculosis coinfection have a risk of IRIS developing with the initiation of ART, especially with early initiation.²² The risk of IRIS led to a debate over the appropriate timing of ART initiation in patients with HIV and tuberculosis coinfection. Numerous studies have shown a survival benefit with the early initiation of ART,²³⁻²⁵ and current guidelines recommend that ART be initiated within 2 weeks after the initiation of antitu-

berculous therapy in patients with HIV and tuberculosis coinfection, except those with tuberculous meningitis.²⁶

This patient had a favorable response to antituberculous therapy, without clinically important side effects or laboratory complications, and the fever promptly resolved. Nine days after the initiation of antituberculous therapy, ART with emtricitabine, tenofovir disoproxil fumarate, and dolutegravir was started.

One week later, while the course of dexamethasone was being tapered, fever with a high temperature recurred, prompting concerns for resistant tuberculosis, an additional undiagnosed opportunistic infection, an adverse drug reaction, or IRIS. No additional infectious source of fever or evidence of drug hypersensitivity was identified on extensive evaluation, which made IRIS the presumptive explanation for the fever. The HIV RNA viral load obtained 12 days after the initiation of ART was 113 copies per milliliter, indicating a robust virologic response, which is a known risk factor for IRIS.²⁷ Glucocorticoid therapy was reinitiated and resulted in prompt resolution of the fever.

FOLLOW-UP

Dr. Park: The patient was discharged to a medical respite facility after 33 days of treatment with antituberculous therapy, ART, and a tapering course of glucocorticoids. At a follow-up visit with her infectious disease physician 3 months after discharge, antituberculous therapy and ART were continued; the CD4+ T-cell count was 94 per microliter, and the HIV RNA viral load was undetectable. Three months after discharge from the hospital, the patient was again hospitalized for depression and suicidality in the context of housing concerns. She has since completed antituberculous therapy, has continued to have viral suppression, and is in early recovery from stimulant use disorder.

FINAL DIAGNOSIS

Disseminated *Mycobacterium tuberculosis* infection.

This case was presented at Medicine Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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REFERENCES

- Meintjies G, Maartens G. HIV-associated tuberculosis. *N Engl J Med* 2024;391:343-55.
- Case Records of the Massachusetts General Hospital (Case 31-2013). *N Engl J Med* 2013;369:1453-61.
- Morris DG, Jasmer RM, Huang L, Gotway MB, Nishimura S, King TE Jr. Sarcoidosis following HIV infection: evidence for CD4+ lymphocyte dependence. *Chest* 2003;124:929-35.
- Eisner MD, Kaplan LD, Herndier B, Stulberg MS. The pulmonary manifestations of AIDS-related non-Hodgkin's lymphoma. *Chest* 1996;110:729-36.
- Edinburgh KJ, Jasmer RM, Huang L, et al. Multiple pulmonary nodules in AIDS: usefulness of CT in distinguishing among potential causes. *Radiology* 2000;214:427-32.
- Digumarthy SR, Mendoza DP, Padole A, et al. Diffuse lung metastases in EGFR-mutant non-small cell lung cancer. *Cancers (Basel)* 2019;11:1360.
- Barlow M, Hamilton W, Ukoumunne OC, Bailey SER. The association between thrombocytosis and subtype of lung cancer: a systematic review and meta-analysis. *Transl Cancer Res* 2021;10:1249-60.
- Salahuddin M, Karanth S, Ocazionez D, Estrada-Y-Martin RM, Cherian SV. Clinical characteristics and etiologies of miliary nodules in the US: a single-center study. *Am J Med* 2019;132:767-9.
- Wheat LJ, Connolly-Stringfield PA, Baker RL, et al. Disseminated histoplasmosis in the acquired immune deficiency syndrome: clinical findings, diagnosis and treatment, and review of the literature. *Medicine (Baltimore)* 1990;69:361-74.
- Falci DR, Monteiro AA, Braz Caurio CE, et al. Histoplasmosis, an underdiagnosed disease affecting people living with HIV/AIDS in Brazil: results of a multicenter prospective cohort study using both classical mycology tests and *histoplasma* urine antigen detection. *Open Forum Infect Dis* 2019;6(4):ofz073.
- Hepler SA, Kaufeld KA, Benedict K, et al. Integrating public health surveillance and environmental data to model presence of histoplasma in the United States. *Epidemiology* 2022;33:654-9.
- Buchacz K, Lau B, Jing Y, et al. Incidence of AIDS-defining opportunistic infections in a multicohort analysis of HIV-infected persons in the United States and Canada, 2000-2010. *J Infect Dis* 2016;214:862-72.
- Wills NK, Adriaanse M, Erasmus S, Wasserman S. Chest X-ray features of HIV-associated *pneumocystis* pneumonia (PCP) in adults: a systematic review and meta-analysis. *Open Forum Infect Dis* 2024;11(4):ofae146.
- Quist J, Hill AR. Serum lactate dehydrogenase (LDH) in *Pneumocystis carinii* pneumonia, tuberculosis, and bacterial pneumonia. *Chest* 1995;108:415-8.
- Canueto-Quintero J, Caballero-Granado FJ, Herrero-Romero M, et al. Epidemiological, clinical, and prognostic differences between the diseases caused by *Mycobacterium kansasii* and *Mycobacterium tuberculosis* in patients infected with human immunodeficiency virus: a multicenter study. *Clin Infect Dis* 2003;37:584-90.
- Post FA, Wood R, Pillay GP. Pulmonary tuberculosis in HIV infection: radiographic appearance is related to CD4+ T-lymphocyte count. *Tuber Lung Dis* 1995;76:518-21.
- Chamie G, Luetkemeyer A, Walusimbi-Nanteza M, et al. Significant variation in presentation of pulmonary tuberculosis across a high resolution of CD4 strata. *Int J Tuberc Lung Dis* 2010;14:1295-302.
- Jones BE, Young SM, Antoniskis D, Davidson PT, Kramer F, Barnes PF. Relationship of the manifestations of tuberculosis to CD4 cell counts in patients with human immunodeficiency virus infection. *Am Rev Respir Dis* 1993;148:1292-7.
- Sharma SK, Mohan A, Sharma A. Miliary tuberculosis: a new look at an old foe. *J Clin Tuberc Other Mycobact Dis* 2016;3:13-27.
- Ray M, Logan R, Sterne JA, et al. The effect of combined antiretroviral therapy on the overall mortality of HIV-infected individuals. *AIDS* 2010;24:123-37.
- Zolopa A, Andersen J, Powderly W, et al. Early antiretroviral therapy reduces AIDS progression/death in individuals with acute opportunistic infections: a multicenter randomized strategy trial. *PLoS One* 2009;4(5):e5575.
- Abdool Karim SS, Naidoo K, Grobler A, et al. Integration of antiretroviral therapy with tuberculosis treatment. *N Engl J Med* 2011;365:1492-501.
- Abdool Karim SS, Naidoo K, Grobler A, et al. Timing of initiation of antiretroviral drugs during tuberculosis therapy. *N Engl J Med* 2010;362:697-706.
- Havir DV, Kendall MA, Ive P, et al. Timing of antiretroviral therapy for HIV-1 infection and tuberculosis. *N Engl J Med* 2011;365:1482-91.
- Blanc F-X, Sok T, Laureillard D, et al. Earlier versus later start of antiretroviral therapy in HIV-infected adults with tuberculosis. *N Engl J Med* 2011;365:1471-81.
- Gandhi RT, Landovitz RJ, Sax PE, et al. Antiretroviral drugs for treatment and prevention of HIV in adults: 2024 recommendations of the International Antiviral Society—USA panel. *JAMA* 2025;333:609-28.
- Grant PM, Komarow L, Andersen J, et al. Risk factor analyses for immune reconstitution inflammatory syndrome in a randomized study of early vs. deferred ART during an opportunistic infection. *PLoS One* 2010;5(7):e11416.

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